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EPIDEMIOLOGY

Lack of association between *CYP17* MspA1 polymorphism and breast cancer risk: a meta-analysis of 22,090 cases and 28,498 controls

Chen Mao · Xi-Wen Wang · Ben-Fu He · Li-Xin Qiu · Ru-Yan Liao · Rong-Cheng Luo · Qing Chen

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Abstract Epidemiological studies have evaluated the association between *CYP17* MspA1 polymorphism and breast cancer risk. However, the results remain conflicting rather than conclusive. In order to derive a more precise estimation of the relationship, we performed the meta-analysis proposed in this article. Systematic searches of the PubMed and Medline databases were performed. A total of 35 studies including 22,090 cases and 28,498 controls were identified. Genotype distributions of *CYP17* in the controls of all studies were in agreement with Hardy–Weinberg equilibrium (HWE) except for three studies. When all the 35 studies were pooled into the meta-analysis, there was no evidence for significant association between *CYP17* MspA1 polymorphism and breast cancer risk (for A1/A2 vs. A1/A1: OR = 1.00, 95% CI = 0.96–1.04; for A2/A2 vs. A1/A1: OR = 1.03, 95% CI = 0.97–1.08; for dominant model: OR = 1.01, 95% CI = 0.97–1.05; for recessive model: OR = 1.03, 95% CI = 0.98–1.08). In the subgroup analyses by ethnicity, menopausal status, and source of controls, no significant associations were found in all genetic models. When sensitivity analyses were performed by excluding HWE-violating studies, all the results were not materially altered. In summary, the meta-analysis strongly suggests that *CYP17* MspA1 polymorphism is not associated with increased breast cancer risk.

Keywords *CYP17* · Polymorphism · Breast cancer · Susceptibility · Meta-analysis

Chen Mao and Xi-Wen Wang contributed equally to this work and should be considered as co-first authors.

C. Mao · X.-W. Wang · R.-Y. Liao · Q. Chen (✉)
Department of Epidemiology, School of Public Health and Tropical Medicine, Southern Medical University, Guangzhou, China
e-mail: epidemiology2008@yahoo.cn

B.-F. He · R.-C. Luo
Department of Oncology, Nanfang Hospital, Southern Medical University, Guangzhou, China

L.-X. Qiu
Department of Medical Oncology, Cancer Hospital, Fudan University, Shanghai, China

L.-X. Qiu
Department of Oncology, Shanghai Medical College, Fudan University, Shanghai, China

Introduction

Cytochrome P450C17α (CYP17) enzyme catalyzes both steroid 17α-hydroxylase and 17,20-lyase activities and functions at key branch points in estrogen biosynthesis [1]. Estrogen has been reported to contribute to the development of cancer via DNA damage either directly or through its metabolites [2, 3]. Therefore, it is conceivable that changes in the activity of CYP17 enzyme may have an impact on estrogen biosynthesis and, thus, influence susceptibility to breast cancer risk.

The *CYP17* gene maps to chromosome 10q24.3 [4]. The 5′-untranslated region of *CYP17* contains a single T to C base transition at position 1931 (rs743572) that creates a recognition site for the MspA1 restriction enzyme and has been used to designate two alleles: A1 (wild allele) and A2 (variant allele) [5]. This polymorphism generates an additional Sp1-type (CCACC box) promoter site (designated A2 allele) 34-bp upstream from the initiation of translation but downstream from the transcription start site [5]. This additional promoter site may enhance transcription of the *CYP17* gene and, thus, increase enzyme activity [6]. Since the variant A2 allele is located in the promoter region, it is biologically reasonable to hypothesize that women who carry variant *CYP17*-A2 allele may have higher breast cancer risk.

To date, many epidemiological studies have evaluated the association between *CYP17* MspA1 polymorphism and breast cancer risk [7–47], but the results remain conflicting rather than conclusive. Ye et al. [48] concluded that there was no evidence for significant association between *CYP17* MspA1 polymorphism and breast cancer risk in a meta-analysis of 15 case–control studies. However, when they performed the meta-analysis, the pooled sample size is relatively small. As some new studies with a large sample-size are emerging, in order to address a more precise estimation of relationship, a meta-analysis including a total of 35 studies were performed, which may provide the most comprehensive evidence for association of *CYP17*-A2 allele with breast cancer risk.

Materials and methods

Publication search

Systematic searches of the PubMed and Medline databases (up to November 30, 2009) were performed using the following search terms: "Cytochrome P450c17α", "*CYP17*", "polymorphism", "mutation", "variant", and "breast cancer". The search was limited to human studies. All the eligible studies were retrieved, and their bibliographies were checked for other relevant publications. When the same patient population was used in several publications, only the most recent, largest or complete study was included in this meta-analysis.

Inclusion criteria

The following criteria were used for the study selection: (1) evaluation of the association between *CYP17* MspA1 polymorphism and breast cancer risk; (2) case–control studies; (3) studies with full text articles; (3) sufficient data for estimating an odds ratio (OR) with 95% confidence interval (CI).

Data extraction

Information was carefully extracted from all the eligible publications independently by two of the authors according to the inclusion criteria listed above. Disagreement was resolved by discussion between the two authors. If these two authors could not reach a consensus, then another author was consulted to resolve the dispute and a final decision was made by the majority of the votes. The following data were collected from each study: first author's name, publication date, country, ethnicity, source of controls, total numbers of cases and controls, and numbers of every genotype, in the respective order. Different descents were categorized as Caucasian, Asian, and African.

Statistical methods

The strength of association between *CYP17* MspA1 polymorphism and breast cancer risk was measured by ORs with 95% CIs. The pooled ORs were estimated for codominant model (A1/A2 vs. A1/A1; A2/A2 vs. A1/A1), dominant model (A1/A2 + A2/A2 vs. A1/A1), and recessive model (A2/A2 vs. A1/A2 + A1/A1). Heterogeneity assumption was checked by the chi-square-based Q-test [49]. A *P* value greater than 0.10 for the Q-test indicates a lack of heterogeneity among studies, and so the fixed-effects model was used for the meta-analysis [50]. Otherwise, the random-effects model was used [51]. Subgroup analyses were performed by ethnicity, menopausal status, and source of controls. Sensitivity analyses were carried out by limiting the meta-analysis to studies conforming to Hardy–Weinberg equilibrium (HWE). An estimate of potential publication bias was carried out by the funnel plot, in which the standard error of log (OR) of each study was plotted against its log (OR). An asymmetric plot suggests a possible publication bias. Funnel plot asymmetry was assessed by the method of Egger's linear regression test, a linear regression approach to measure funnel plot asymmetry on the natural logarithm scale of the OR. The significance of the intercept was determined by the *t*-test suggested by Egger ($P < 0.05$ was considered representative of statistically significant publication bias) [52]. All of the statistical tests used in our meta-analysis were performed by STATA version 10.0 (Stata Corporation, College Station, TX).

Results

Study characteristics

Based on our search criteria, a total of 41 studies were preliminarily eligible [7–47]. Among them, six studies [7, 20, 22, 30, 32, 37] were excluded because the same data were available in other studies [21, 23, 33, 35, 36]. Thus, 35 studies were included in the final meta-analysis. Table 1 lists the studies thus identified and their main characteristics. The study of Young et al. [15] and the study of Gudmundsdottir et al. [24] included the data on both male and female breast cancer; however, only data on female breast cancer were included in the pooled analysis. In total, these studies included 22,090 cases and 28,498 controls. There were 18 studies of Caucasians, 13 studies of Asians, and 4 studies of mixed populations. Genotype distributions in the controls of all studies were in agreement with HWE except for three studies [16, 17, 36].

Table 1 Main characteristics of studies included in the meta-analysis

First author	Year	Country	Ethnicity	Source of controls	Genotypes distribution (case/control)			HWE
					A1/A1	A1/A2	A2/A2	
Dunning [8]	1998	UK	Caucasian	PB	303/229	402/277	130/85	0.93
Helzlsouer [9]	1998	USA	Caucasian	PB	41/37	47/58	21/18	0.55
Weston [10]	1998	USA	Mixed	HB	45/92	57/111	21/37	0.71
Bergman-Jungeström [11]	1999	Sweden	Caucasian	HB	32/53	62/55	15/9	0.30
Haiman [12]	1999	USA	Caucasian	PB	178/217	212/307	73/94	0.39
Huang [13]	1999	Taiwan	Asian	HB	25/28	54/63	44/35	0.97
Nedelcheva Kristensen [14]	1999	Norway	Caucasian	PB	202/74	241/101	67/26	0.35
Young [15]	1999	UK	Caucasian	PB	21/23	13/28	5/7	0.73
Hamajima [16]	2000	Japan	Asian	HB	41/44	83/95	20/27	0.04
Kuligina [17]	2000	Russia	Caucasian	HB	82/61	101/77	47/44	0.05
Mitrunen [18]	2000	Finland	Caucasian	PB	199/200	227/220	53/60	0.97
Miyoshi [19]	2000	Japan	Asian	HB	178/154(A1/A1 + A1/A2)		61/41	0.69
Feigelson [21]	2001	USA	Mixed	PB	292/542	409/739	149/227	0.34
Ambrosone [23]	2003	USA	Caucasian	PB	109/95	83/71	15/22	0.13
Gudmundsdottir [24]	2003	Iceland	Caucasian	PB	193/156	247/173	60/66	0.13
Tan [25]	2003	China	Asian	PB	46/51	115/110	89/89	0.12
Wu [26]	2003	Singapore	Asian	PB	37/109	82/333	69/229	0.51
Ahsan [27]	2004	USA	Mixed	FB	59/56	87/98	25/34	0.43
Hefler [28]	2004	Austria	Caucasian	PB	127/607	186/804	75/287	0.45
Chacko [29]	2005	India	Asian	HB	94/115	40/22	6/3	0.13
Einarsdóttir [31]	2005	Sweden	Caucasian	PB	550/488	711/638	238/212	0.88
Hopper [33]	2005	Australia	Caucasian	PB	553/311	621/364	230/113	0.70
Shin [34]	2005	Korean	Asian	HB	112/70	223/152	127/115	0.13
Verla-Tebit [35]	2005	Germany	Caucasian	PB	180/323	242/424	103/157	0.38
Hu [36]	2006	China	Asian	HB	53/100	105/235	52/92	0.04
Chakraborty [38]	2007	India	Asian	PB	29/57	98/110	59/45	0.55
Setiawan [39]	2007	USA	Mixed	PB	1869/2474	2445/3338	833/1070	0.31
Chen [40]	2008	USA	Caucasian	PB	363/398	506/523	168/175	0.88
Sakoda [41]	2008	China	Asian	PB	102/138	297/441	216/298	0.23
Torresan [42]	2008	Brazil	Caucasian	PB	35/46	61/48	6/8	0.35
Zhang [43]	2008	China	Asian	PB	47/69	169/208	84/113	0.11
Antognelli [44]	2009	Italy	Caucasian	PB	229/227	258/249	60/68	0.98
MARIE-GENICA [45]	2009	Germany	Caucasian	PB	1043/1834	1573/2712	529/941	0.25
Samson [46]	2009	India	Asian	NR	127/220	91/226	32/54	0.72
Sangrajrang [47]	2009	Thailand	Asian	HB	187/167	281/230	96/92	0.42

FB family-based, *HB* hospital-based, *HWE* Hardy–Weinberg equilibrium, *PB* population-based, *NR* not reported

Quantitative synthesis

Table 2 lists the main results of the meta-analysis. When all the 35 studies were pooled into the meta-analysis, there was no evidence for significant association between *CYP17* MspA1 polymorphism and breast cancer risk (for A1/A2 vs. A1/A1: OR = 1.00, 95% CI = 0.96–1.04; for A2/A2 vs. A1/A1: OR = 1.03, 95% CI = 0.97–1.08; for dominant model: OR = 1.01, 95% CI = 0.97–1.05; for recessive model: OR = 1.03, 95% CI = 0.98–1.08) (Table 2). In the subgroup analyses by ethnicity, menopausal status, and source of controls, no significant associations were found in all genetic models (Table 2).

Sensitivity analyses

Sensitivity analyses were conducted to ascertain whether modification of the inclusion criteria of the meta-analysis affected the final results. These were carried out by limiting the meta-analysis to studies conforming to HWE. All the

Table 2 Results of meta-analysis for *CYP17* MspAI polymorphism and breast cancer risk

Study groups	No. of study	Sample size (case/control)	A1/A2 vs. A1/A1		A2/A2 vs. A1/A1		A1/A2 + A2/A2 vs. A1/A1		A2/A2 vs. A1/A2 + A1/A1	
			OR (95% CI)	P_h	OR (95% CI)	P_h	OR (95% CI)	P_h	OR (95% CI)	P_h
Overall	35	22,090/28,498	1.00 (0.96–1.04)	0.24	1.03 (0.97–1.08)	0.34	1.01 (0.97–1.05)	0.17	1.03 (0.98–1.08)	0.41
Ethnicity										
Asian	13	3,671/4,780	1.01 (0.86–1.19)	0.03	1.02 (0.89–1.17)	0.11	1.03 (0.88–1.22)	0.02	1.05 (0.94–1.16)	0.19
Caucasian	18	12,128/14,900	1.02 (0.97–1.08)	0.54	1.01 (0.94–1.09)	0.57	1.02 (0.97–1.07)	0.55	1.01 (0.94–1.08)	0.55
Mixed	4	6,291/8,818	0.98 (0.91–1.05)	0.85	1.05 (0.95–1.16)	0.37	0.99 (0.93–1.06)	0.62	1.06 (0.97–1.16)	0.49
Menopausal status										
Premenopausal	14	3,203/3,450	0.98 (0.87–1.09)	0.52	1.00 (0.86–1.17)	0.69	0.97 (0.87–1.07)	0.49	1.00 (0.87–1.14)	0.63
Postmenopausal	15	4,689/5,067	0.98 (0.89–1.08)	0.52	1.04 (0.91–1.18)	0.94	0.99 (0.91–1.08)	0.71	1.06 (0.94–1.18)	0.90
Source of controls										
Hospital-based	10	2,344/2,419	0.08 (0.93–1.24)	0.14	0.96 (0.80–1.15)	0.16	1.09 (0.90–1.33)	0.04	0.99 (0.85–1.14)	0.17
Population-based	23	19,325/25,391	1.00 (0.96–1.05)	0.58	1.04 (0.98–1.10)	0.47	1.01 (0.97–1.05)	0.59	1.04 (0.99–1.09)	0.55

Study groups	A1/A2 + A2/A2 vs. A1/A1		A2/A2 vs. A1/A2 + A1/A1	
	OR (95% CI)	P_h	OR (95% CI)	P_h
Overall	1.01 (0.97–1.05)	0.17	1.03 (0.98–1.08)	0.41
Ethnicity				
Asian	1.03 (0.88–1.22)	0.02	1.05 (0.94–1.16)	0.19
Caucasian	1.02 (0.97–1.07)	0.55	1.01 (0.94–1.08)	0.55
Mixed	0.99 (0.93–1.06)	0.62	1.06 (0.97–1.16)	0.49
Menopausal status				
Premenopausal	0.97 (0.87–1.07)	0.49	1.00 (0.87–1.14)	0.63
Postmenopausal	0.99 (0.91–1.08)	0.71	1.06 (0.94–1.18)	0.90
Source of controls				
Hospital-based	1.09 (0.90–1.33)	0.04	0.99 (0.85–1.14)	0.17
Population-based	1.01 (0.97–1.05)	0.59	1.04 (0.99–1.09)	0.55

P_h *P* values for heterogeneity from Q-test

results were not materially altered and did not draw different conclusions (data not shown).

Publication bias

We performed Begg's funnel plot and Egger's test to assess the publication bias of literatures. The shapes of the funnel plots did not reveal any evidence of obvious asymmetry. We present funnel plot for ORs of A1/A2 vs. A1/A1 in Fig. 1. Also, the results of Egger's test still did not suggest any evidence of publication bias ($P = 0.48$ for A1/A2 vs. A1/A1, $P = 0.81$ for A2/A2 vs. A1/A1, $P = 0.45$ for dominant model and $P = 0.82$ for recessive model, respectively).

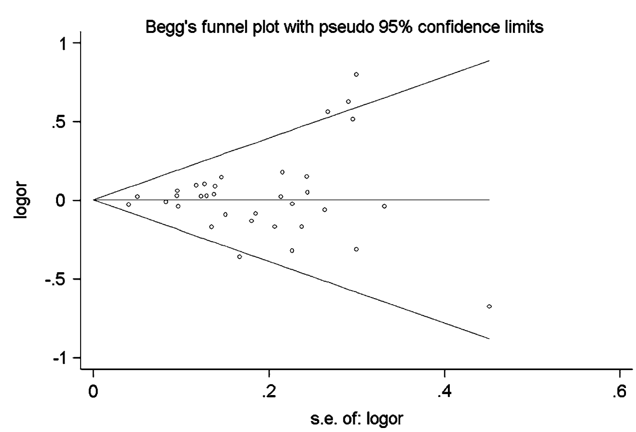

Fig. 1 Funnel plot analysis for odds ratios of A1/A2 genotype compared with A1/A1 genotype in overall studies

Discussion

It has been hypothesized that women who carry *CYP17*-A2 allele may exhibit increased activity of CYP17 enzyme [6]. Increased CYP17 activity may have a positive impact on estrogen biosynthesis, and, therefore, cause a higher risk of breast cancer. To date, a number of epidemiological studies have evaluated the association between *CYP17* MspA1 polymorphism and breast cancer risk, but the results remain inconclusive. In order to derive a more precise estimation of relationship, we performed this meta-analysis. The present meta-analysis of 35 studies, including 22,090 cases and 28,498 controls, provide most comprehensive analysis on the association of *CYP17* MspA1 polymorphism with breast cancer risk. The results indicated that the *CYP17* MspA1 polymorphism is not associated with elevated breast cancer risk in the overall studies population. These findings were consistent with most of the related studies as summarized in our meta-analysis.

It has been suggested that the role of *CYP17* MspA1 polymorphism in breast cancer development may mediate by ethnicity and menopausal status. However, when we conduct subgroup analyses according to different ethnicities (Asians and Caucasians) and different menopausal status (premenopausal and postmenopausal), no significant association was found in any subgroup of population.

Heterogeneity is a potential problem that may affect the interpretation of the results. Despite some diversity in the studies about study designs, sample sizes, inclusion criteria, ethnicity, and menopausal status, there was no statistically significant heterogeneity in overall comparisons. This indicated that it may be appropriate to use an overall estimation of the relationship between *CYP17* MspA1 polymorphism and breast cancer risk.

Some limitations of this meta-analysis should be acknowledged. First, some studies found significant associations between *CYP17* MspA1 polymorphism and breast cancer risk in several subgroups of populations, such as associations among postmenopausal women with high body mass index (BMI) [40], patients with advanced breast cancer [7], or women at young ages [11, 20, 34, 38]. It is difficult for a meta-analysis to derive such specific associations because the results from previous studies were not presented in a uniform standard. Second, our result was based on single-factor estimates without adjustment for other risk factors such as age at menarche, BMI, ethnicity, family history of breast caner, and environmental factors. Despite its limitations, our meta-analysis also had some advantages. First, substantial number of cases and controls were pooled from different studies, which significantly increased the statistical power of the analysis. Second, the results of subgroup analyses and sensitivity analyses were not materially altered and did not draw different conclusions, indicating that our results were robust. Third, Begg's and Egger's tests did not detect any publication bias, indicating that our results were unbiased.

In conclusion, this meta-analysis of 35 studies strongly suggests that *CYP17* MspA1 polymorphism is not associated with increased breast cancer risk.

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